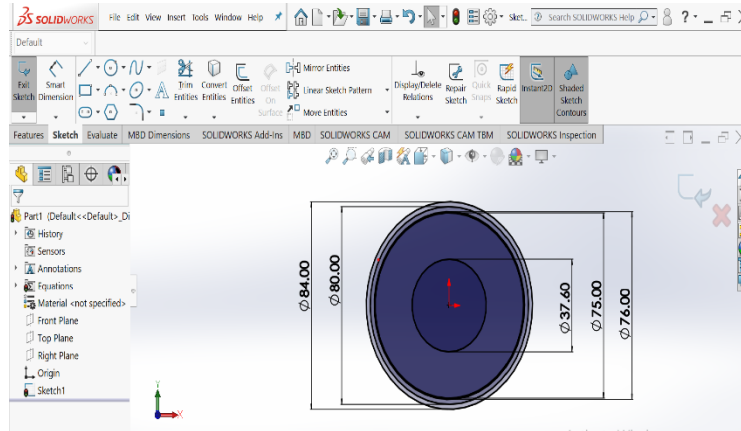


GEAR DESIGN ON SOLID WORKS

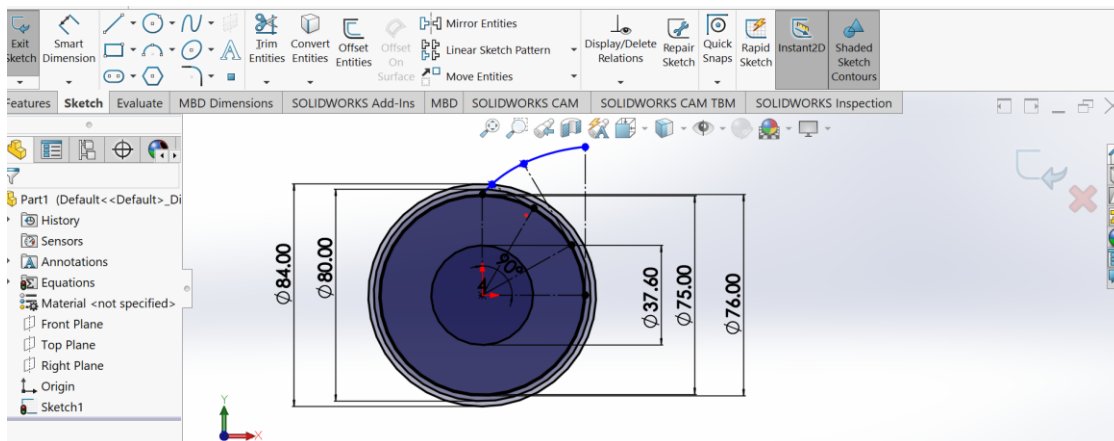
- ***NAME : Owais Pervaiz***
- ***ROLL NO : FA22-
BME-035***
- ***COURSE TITLE :
MACHINE DESIGN II***
- ***SUBMITTED TO :
ENGR.HAROON
MUSHTAQ***



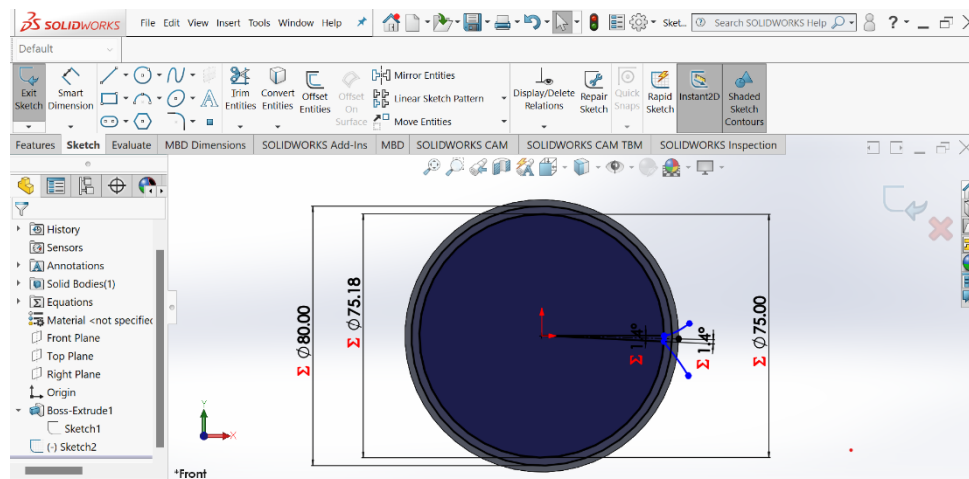
Sketch:



Involute profile:



Involute Profile by using Equations of machine design



$$= \sqrt{("dp" ^ 2 - "db" ^ 2) / "db" * 180 / \pi - "phi"}$$

$$"D2@Sketch2"*0.5*(\cos(-t-2*"D4@Sketch2"*\pi/180)-t*\sin(-t-2*"D4@Sketch2"*\pi/180))$$

$$"D2@Sketch2"*0.5*(\cos(-t-2*"D4@Sketch2"*\pi/180)-t*\sin(-t-2*"D4@Sketch2"*\pi/180))$$

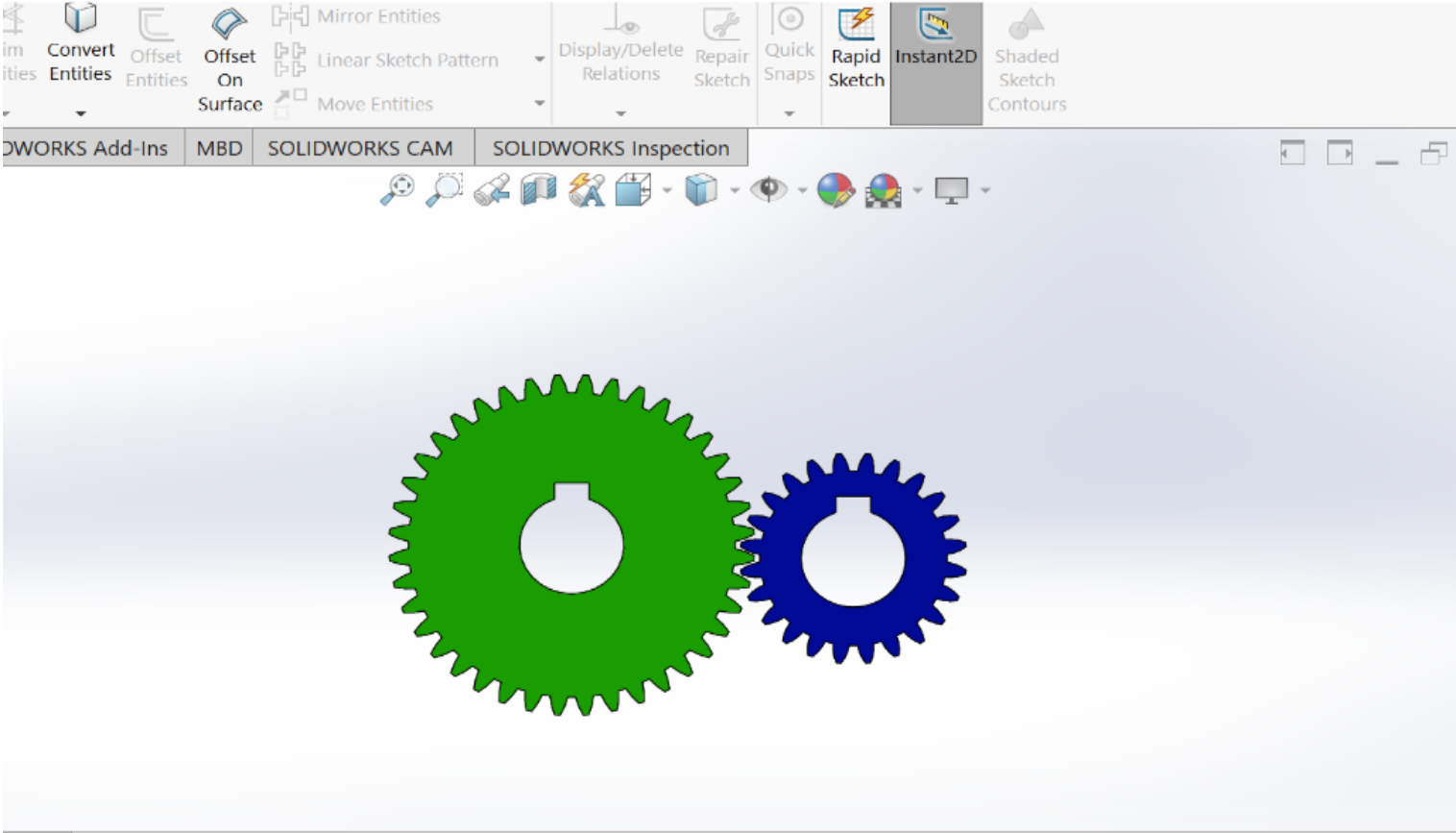
$$"D2@Sketch3"*0.5*(\cos(t)+t*\sin(t))$$

after using these equations we have got the above profile

Pressure angle: 14.5 Equations

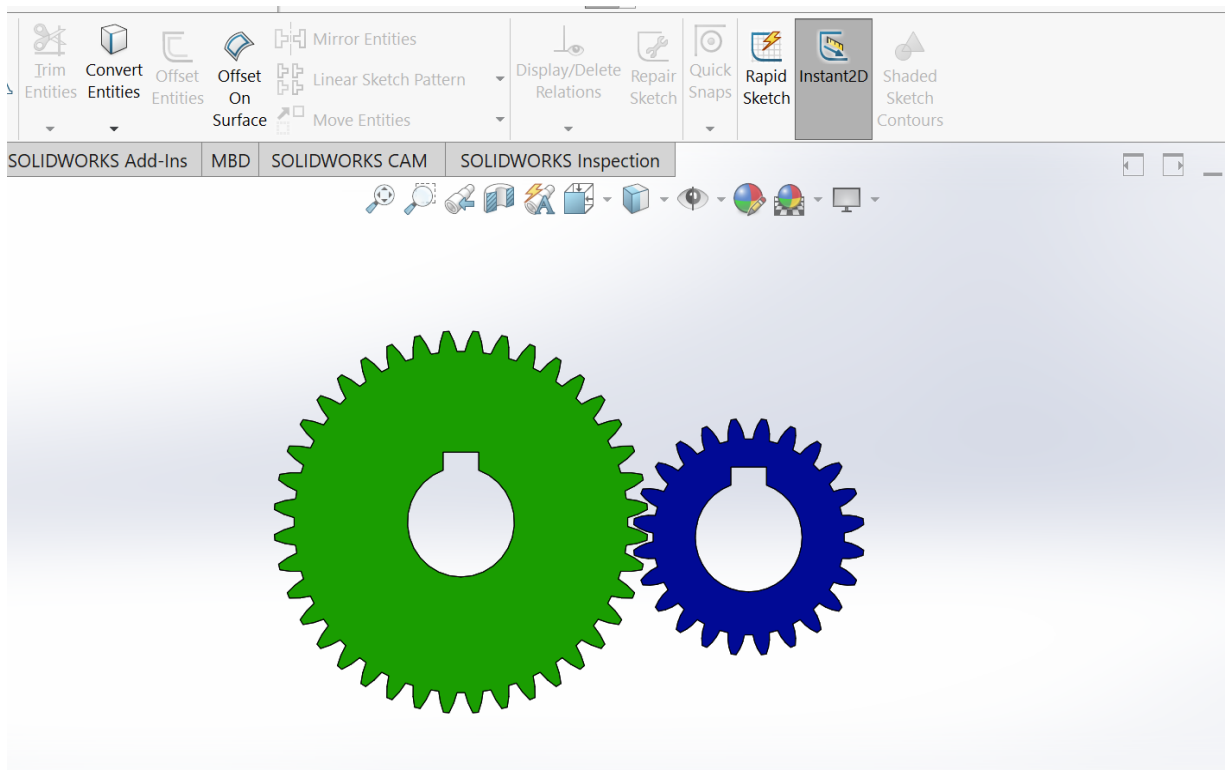
Equations, Global Variables, and Dimensions

Name	Value / Equation	Evaluates to	Comments
Global Variables			
"p"	= 0.5mm	0.5mm	diametral pitch
"n"	= 40	40	number of teeth
"phi"	= 14.5deg	14.5deg	pressure angle
"a"	= 1 / "p"	2mm	addendum
"b"	= 1.25 / "p"	2.5mm	dedendum
"c"	= "b" - "a"	0.5mm	clearance
"dp"	= "n" / "p"	80mm	pitch circle diameter
"db"	= "dp" * cos ("phi")	77.4518deg	base circle diameter
Add global variable			
Features			
Add feature suppression			
Equations			
"D1@Sketch2"	= "dp" + 2 * "a"	84mm	addendum circle diameter
"D1@Sketch3"	= "dp"	80mm	pitch circle dia
"D2@Sketch3"	= "db"	77.45mm	dase circle dia
"D3@Sketch3"	= "dp" - 2 * "a"	76mm	dendum circle dia
Add equation			



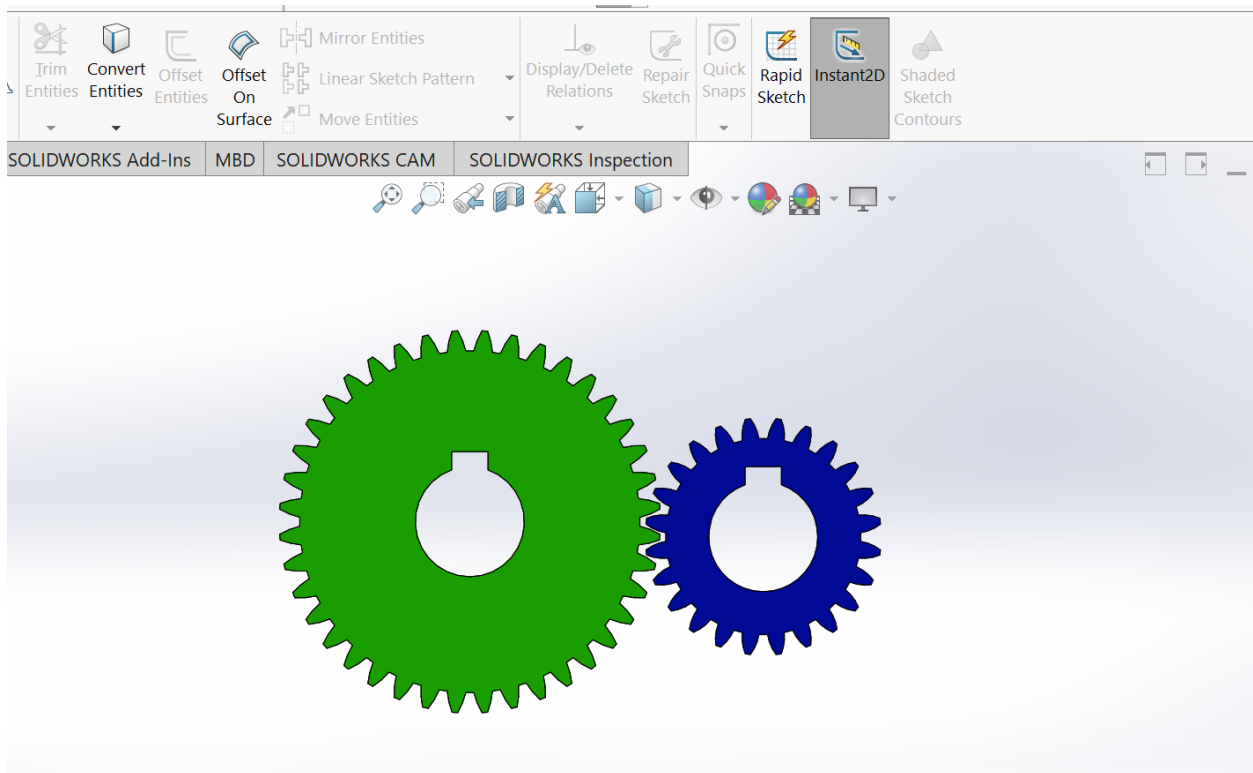
Pressure angle: 20, equations. “Only change pressure angle from equations to create new gears other parameters will automatically change”

Name	Value / Equation	Evaluates to	Comments
Global Variables			
"p"	= 0.5mm	0.5mm	diametral pitch
"n"	= 40	40	number of teeth
"phi"	= 20deg	20deg	pressure angle
"a"	= 1 / "p"	2mm	addendum
"b"	= 1.25 / "p"	2.5mm	dedendum
"c"	= "b" - "a"	0.5mm	clearance
"dp"	= "n" / "p"	80mm	pitch circle diameter
"db"	= "dp" * cos ("phi")	75.1754deg	base circle diameter
<i>Add global variable</i>			
Features			
<i>Add feature suppression</i>			
Equations			
"D1@Sketch2"	= "dp" + 2 * "a"	84mm	addendum circle diameter
"D1@Sketch3"	= "dp"	80mm	pitch circle dia
"D2@Sketch3"	= "db"	75.18mm	base circle dia
"D3@Sketch3"	= "dp" - 2 * "a"	76mm	dedendum circle dia
<i>Add equation</i>			



Pressure angle: 22, Equations.

Name	Value / Equation	Evaluates to	Comments
Global Variables			
"p"	= 0.5mm	0.5mm	diametral pitch
"n"	= 40	40	number of teeth
"phi"	= 22deg	22deg	pressure angle
"a"	= 1 / "p"	2mm	addendum
"b"	= 1.25 / "p"	2.5mm	dedendum
"c"	= "b" - "a"	0.5mm	clearance
"dp"	= "n" / "p"	80mm	pitch circle diameter
"db"	= "dp" * cos ("phi")	74.1747deg	base circle diameter
<i>Add global variable</i>			
Features			
<i>Add feature suppression</i>			
Equations			
"D1@Sketch2"	= "dp" + 2 * "a"	84mm	addendum circle diameter
"D1@Sketch3"	= "dp"	80mm	pitch circle dia
"D2@Sketch3"	= "db"	74.17mm	base circle dia
"D3@Sketch3"	= "dp" - 2 * "a"	76mm	dedendum circle dia
<i>Add equation</i>			



Name	Value / Equation	Evaluates to	Comments
Global Variables			
"p"	= 0.5mm	0.5mm	diametral pitch
"n"	= 40	40	number of teeth
"phi"	= 24deg	24deg	pressure angle
"a"	= 1 / "p"	2mm	addendum
"b"	= 1.25 / "p"	2.5mm	dedendum
"c"	= "b" - "a"	✓ 0.5mm	clearance
"dp"	= "n" / "p"	80mm	pitch circle diameter
"db"	= "dp" * cos ("phi")	73.0836deg	base circle diameter
<i>Add global variable</i>			
Features			
<i>Add feature suppression</i>			
Equations			
"D1@Sketch2"	= "dp" + 2 * "a"	84mm	addendum circle diameter
"D1@Sketch3"	= "dp"	80mm	pitch circle dia
"D2@Sketch3"	= "db"	73.08mm	base circle dia
"D3@Sketch3"	= "dp" - 2 * "a"	76mm	dedendum circle dia
<i>Add equation</i>			

